

ELTE IK - Computer Science BSc

Final Exam Topics

4. Number theory, graphs

Note: This document was translated from Hungarian to English by AI. Please verify the information against the original Hungarian source before relying on it.

Number theory, graphs, coding theory

Sets, relations, functions and operations. Complex numbers. Enumerations on finite sets. Undirected and directed graphs, trees, Eulerian and Hamiltonian graphs, data structures of graphs. Basic concepts of number theory, divisibility, congruence, primes. Polynomials and their operations, division with remainder.

1 Number theory

1.1 Sets

A set (system, class, collection, ...) is the conceptual envelope of its elements. A set is uniquely determined by its elements.

Basic concepts

- Empty set

The set that has no elements is the empty set. Notation: $\emptyset$. Based on the axiom of determinacy this is unique

- Subset

We say that **A is a subset of B** ($A \subseteq B$) if $\forall a \in A : a \in B$, that is, B contains every element of A. **A is a proper subset of B** $A \subset B$ if $A \subseteq B$, but $A \neq B$, that is, B has at least one element that is not an element of A.

- Power set

If A is a set, then the set system whose elements are exactly the subsets of the set A is called the power set of A , and is denoted by 2^A .

- $A = \emptyset, 2^\emptyset = \{\emptyset\}$
- $A = \{a\}, 2^{\{a\}} = \{\emptyset, \{a\}\}$
- $A = \{a, b\}, 2^{\{a, b\}} = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$
- $|2^A| = 2^{|A|}$

Operations

- Union

The union of the sets A and B : $A \cup B$ is the set that contains exactly the elements of A and B .

- Intersection

The intersection of the sets A and B : $A \cap B$ is the set that contains exactly the common elements of A and B : $A \cap B = \{x \in A : x \in B\}$

- Disjoint
If $A \cap B = \emptyset$, then A and B are disjoint.
- Difference
The difference of the sets A and B is $A \setminus B = \{x \in A : x \notin B\}$
- Complement
For a fixed universe X and a subset $A \subseteq X$, the complement of the set A is $\bar{A} = A' = X \setminus A$
- Symmetric difference
 $A \Delta B = (A \setminus B) \cup (B \setminus A)$

Properties

- Union
 - $A \cup \emptyset = A$
 - $A \cup (B \cup C) = (A \cup B) \cup C$ (associativity)
 - $A \cup B = B \cup A$ (commutativity)
 - $A \cup A = A$ (idempotence)
 - $A \subseteq B \iff A \cup B = B$
- Intersection
 - $A \cap \emptyset = \emptyset$
 - $A \cap (B \cap C) = (A \cap B) \cap C$ (associativity)
 - $A \cap B = B \cap A$ (commutativity)
 - $A \cap A = A$ (idempotence)
 - $A \subseteq B \iff A \cap B = A$
- Distributivity properties of union and intersection
 - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
 - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- Difference
 - $A \setminus B = A \cap \bar{B}$
- Complement
 - $\overline{\bar{B}} = B$
 - $\bar{\emptyset} = X$
 - $\bar{X} = \emptyset$
 - $A \cap \bar{A} = \emptyset$
 - $A \cup \bar{A} = X$
 - $A \subseteq B \iff \bar{B} \subseteq \bar{A}$
 - $\overline{A \cap B} = \bar{A} \cup \bar{B}$
 - $\overline{A \cup B} = \bar{A} \cap \bar{B}$
- Symmetric difference
 - $A \Delta B = (A \cup B) \setminus (A \cap B)$

1.2 Relations, orderings

Basic concepts

- Ordered pair
 (x, y) is an ordered pair if $(x, y) = (u, v) \iff x = u \wedge y = v$. We define this property with sets:

$$(x, y) := \{\{x\}, \{x, y\}\}$$

- Cartesian product

The Cartesian product or direct product of the sets X, Y :

$$X \times Y := \{(x, y) : x \in X, y \in Y\}$$

- Binary relation

A set is called a binary relation if every element of it is an ordered pair. If R is a binary relation and $(x, y) \in R$, then we often write: xRy

- Relation

If for sets X, Y we have $R \subset X \times Y$, then R is a relation between X and Y .

- Domain

The domain of the binary relation R :

$$\text{dmn}(R) := \{x \mid \exists y : (x, y) \in R\}$$

- Range

The range of the binary relation R :

$$\text{rng}(R) := \{y \mid \exists x : (x, y) \in R\}$$

- Inverse

The inverse of a binary relation R :

$$R^{-1} := \{(a, b) : (b, a) \in R\}$$

- Image of a set

Let R be a binary relation, and A a set. The image of the set A :

$$R(A) := \{y \mid \exists x \in A : (x, y) \in R\}$$

- Composition

The composition of the binary relations R and S :

$$R \circ S := \{(x, y) \mid \exists z : (x, z) \in S \wedge (z, y) \in R\}$$

Properties

R is a binary relation on X (that is, $R \subset X \times X$)

1. transitive

$$\forall x, y, z : (x, y) \in R \wedge (y, z) \in R \implies (x, z) \in R$$

2. symmetric

$$\forall x, y : (x, y) \in R \implies (y, x) \in R$$

3. antisymmetric

$$\forall x, y : (x, y) \in R \wedge (y, x) \in R \implies x = y$$

4. strictly antisymmetric

$$\forall x, y : (x, y) \in R \implies (y, x) \notin R$$

5. reflexive

$$\forall x \in X : (x, x) \in R$$

6. irreflexive

$$\forall x \in X : (x, x) \notin R$$

7. trichotomous

If for every $x, y \in X$ exactly one of the following holds

- a) $x = y$
- b) $(x, y) \in R$
- c) $(y, x) \in R$

8. dichotomous

$$\forall x, y \in X : (x, y) \in R \vee (y, x) \in R$$

In other words, the elements are comparable.

Orderings

- Equivalence relation, classification

X a set, R a binary relation on X , is an equivalence relation if it is

- Reflexive
- Transitive
- Symmetric

A system $\mathcal{O}$ of subsets of X is called a classification (partition) if $\mathcal{O}$ is a set system consisting of pairwise disjoint nonempty sets for which $\cup \mathcal{O} = X$

Theorem:

An equivalence relation determines a classification. Conversely: for a classification $\mathcal{O}$, $R = \cup \{Y \times Y : Y \in \mathcal{O}\}$ is an equivalence relation.

- Partial order

X a set, R a binary relation on X , is a partial order if it is

- Reflexive
- Transitive
- Antisymmetric

- Total order

X a set, R a binary relation on X , is a (total) order if it is

- Reflexive
- Transitive
- Antisymmetric
- Dichotomous

In plain words, if a partial order is dichotomous (that is, all of its elements are comparable), then it is a (total) order.

- Strict and weak relation, ordering

X a set, R, S relations on X . If

$$xRy \wedge x \neq y \Rightarrow xSy$$

then we call S the strictification of R .

Conversely, if

$$xRy \vee x = y \Rightarrow xTy$$

then T is the weak relation corresponding to R .

Remark: This is essentially about removing and adding reflexivity. For a partial order the corresponding strict relation (strict partial order) is therefore irreflexive, and consequently also strictly antisymmetric. Conversely: The weak relation corresponding to a strict partial order on X (trans., irrefl., strictly antisym.) is a partial order.

Bounds

- Least, greatest, minimal, maximal element

By the least (first) element of a partial order ($\preceq$) on a set X we mean an element $x \in X$ for which: $\forall y \in X : x \preceq y$. (Such an element need not exist, but if it does, it is unique).

Similarly, the greatest (last) element is an $x \in X$ such that $\forall y \in X : y \preceq x$.

We call x minimal if there is no element smaller than it, and maximal if there is no element greater than it. (In contrast to least/greatest elements, there can be several minimal/maximal elements. If, however, X is ordered, then least=minimal, greatest=maximal.)

- Lower bound, upper bound

X a partially ordered set, $Y \subset X$. The element $x \in X$ is a lower bound of Y if $\forall y \in Y : x \preceq y$. (upper bound: $\forall y \in Y : y \preceq x$). It can be seen that x is not necessarily an element of Y ; moreover it is also possible that Y has no lower/upper bound, or even several. If, however, $x \in Y$, then it is unique and it is the least element of Y .

- Infimum, supremum

If among the lower bounds there is a greatest element, it is called the infimum of Y . (Notation: $\inf Y$)

If among the upper bounds there is a least element, it is called the supremum of Y . (Notation: $\sup Y$)

- Lower bound, upper bound property

X a partially ordered set. If $\forall \emptyset \neq Y \subset X : Y$ is bounded above and has a supremum, then it has the upper bound property. Likewise, if $\forall \emptyset \neq Y \subset X : Y$ is bounded below and has an infimum, then it has the lower bound property.

1.3 Functions and operations

1.3.1 Functions

Definition

A relation f is a function if

$$(x, y) \in f \wedge (x, y') \in f \implies y = y'$$

In other words, for every x there exists at most one y such that $(x, y) \in f$

Thus for every $x \in \text{dmn}(f)$ we have $f(x) = \{y\}$, which we also usually denote $f(x) = y$ or $f : x \mapsto y$ or $f_x = y$.

Domain, range

We use the notation $f : X \rightarrow Y$ if $\text{dmn}(f) = X$.

We use the notation $f \in X \rightarrow Y$ if $\text{dmn}(f) \subset X$ (when $\text{dmn}(f) \subsetneq X$ can also occur).

In both cases $\text{rng}(f) \subset Y$.

Injective

a function f is one-to-one/injective if

$$f(x) = y \wedge f(x') = y \implies x = x'$$

This is equivalent to f^{-1} also being a function.

Surjective

The function f is surjective if

$$\forall y \in Y : \exists x \in X : f(x) = y$$

That is, $\text{rng}(f) = Y$. In plain words the function f maps onto the whole of Y .

Bijjective

If the function f is injective and surjective, then it is bijective.

Indexed family

The value of the function x taken at position i is also usually denoted x_i . In this case we often call the domain $\text{dmn}(f) = I$ the index set, its elements indices, $\text{rng}(f)$ the indexed set, and the function x itself an indexed family.

1.3.2 Operations**Definitions**

- Binary operation
On a set X a function $f : X \times X \rightarrow X$ is a binary operation.
- Unary operation
On a set X a function $f : X \rightarrow X$ is a unary operation.
- Nullary operation
 X a set, $f : \{\emptyset\} \rightarrow X$ a nullary operation. (Essentially the selection of an element)

Properties

- Let $\spadesuit, \odot$ be binary operations on X .

1. $\spadesuit$ is associative if

$$\forall x, y, z \in X : (x \spadesuit y) \spadesuit z = x \spadesuit (y \spadesuit z)$$

2. $\spadesuit$ is commutative if

$$\forall x, y \in X : x \spadesuit y = y \spadesuit x$$

3. $\spadesuit$ is distributive over $\odot$ if $\forall x, y, z \in X$:

$$x \spadesuit (y \odot z) = (x \spadesuit y) \odot (x \spadesuit z) \quad \text{- left}$$

$$(y \odot z) \spadesuit x = (y \spadesuit x) \odot (z \spadesuit x) \quad \text{- right}$$

- Let $\heartsuit$ be a binary operation on X and $\S$ a binary operation on Y . $f : X \rightarrow Y$ is operation-preserving if:

$$\forall x_1, x_2 \in X : f(x_1 \heartsuit x_2) = f(x_1) \S f(x_2)$$

1.4 Number concept, complex numbers**1.4.1 Number concept****Algebraic structures**

1. Groupoid

A set G with an operation $\star$, that is, the pair $(G, \star)$, is called a groupoid.

2. Semigroup

If in a groupoid the operation $\star$ is associative, then the groupoid is a semigroup.

3. Monoid

A semigroup with an identity element is called a monoid.

Remark: $a \in G$ is an identity element if $\forall g \in G : a \star g = g \star a = g$

4. Group

If in a monoid every element has an inverse, then we speak of a group.

Remark: $g, g^{-1} \in G$ and $\xi \in G$ is the identity element, then g^{-1} is the inverse of g if $g \star g^{-1} = \xi$ and $g^{-1} \star g = \xi$

5. Abelian group

If in a group the operation is commutative, then it is an Abelian group.

6. Ring

$(R, +, \cdot)$ is a ring if it is an Abelian group under addition, a semigroup under multiplication, and both-sided distributivity holds.

If multiplication is commutative, then it is a commutative ring.

If multiplication has an identity element, then it is a ring with identity.

7. Integral domain

A commutative ring without zero divisors.

Zero divisor: x, y are distinct nonzero elements, but $x \cdot y = 0$

8. Ordered integral domain

An integral domain R is an ordered integral domain if it is an ordered set, and moreover addition and multiplication are monotone.

Addition monotone: $x, y, z \in R$ and $x \leq y \Rightarrow x + z \leq y + z$

Multiplication monotone: $x, y \in R$ and $x, y \geq 0 \Rightarrow x \cdot y \geq 0$

9. Field

A ring R is a field if $R \setminus \{0\}$ is an Abelian group under multiplication.

10. Ordered field

If a field is an ordered integral domain, then it is an ordered field.

Natural numbers

• Peano axioms

Let $\mathbb{N}$ be a set and $+$ a function defined on $\mathbb{N}$. The following conditions are called the Peano axioms:

1. $0 \in \mathbb{N}$ - 0 is a nullary operation on $\mathbb{N}$
2. if $n \in \mathbb{N}$, then $n^+ \in \mathbb{N}$ - $+$ is a unary operation on $\mathbb{N}$
3. if $n \in \mathbb{N}$, then $n^+ \neq 0$ - 0 is not in the range of $+$
4. if $n, m \in \mathbb{N}$, and $m^+ = n^+$, then $n = m$ - $+$ is injective
5. if $S \subset \mathbb{N}$, $0 \in S$, and $n \in S : n^+ \in S$, then $S = \mathbb{N}$ - the principle of mathematical induction

• Operations

– addition

$k, m, n \in \mathbb{N}$, then:

1. $(k + m) + n = k + (m + n)$ - associativity
2. $n + 0 = 0 + n = n$ - 0 is the zero element (additive identity)
3. $n + k = k + n$ - commutativity

4. $n + k = m + k$ or $k + n = k + m$, then $m = n$ - *cancellation rule*
- multiplication
- $k, m, n \in \mathbb{N}$, then:
 1. $(k \cdot m) \cdot n = k \cdot (m \cdot n)$ - *associativity*
 2. $0 \cdot n = n \cdot 0 = 0$
 3. $n \cdot 1 = 1 \cdot n = n$ - *1 is the identity element (multiplicative identity)*
 4. $n \cdot k = k \cdot n$ - *commutativity*
 5. $k \cdot (m + n) = k \cdot m + k \cdot n$, and $(m + n) \cdot k = m \cdot k + n \cdot k$ - *distributivity*
 6. for $k \neq 0$: $n \cdot k = m \cdot k$, then $m = n$ - *cancellation rule*

Integers

In the realm of natural numbers, with respect to addition only zero has an inverse; in other words, subtraction generally cannot be performed.

Consider the relation $\sim \subset \mathbb{N} \times \mathbb{N}$ for which $(m, n) \sim (m', n')$ if $m + n' = m' + n$. And take the addition $(m, n) + (m', n') = (m + m', n + n')$. The relation $\sim$ is an equivalence relation; we denote the set of equivalence classes by $\mathbb{Z}$. We call the elements of $\mathbb{Z}$ integers.

Addition is compatible with the equivalence, so it is defined among the integers, and $(\mathbb{Z}, +)$ is an Abelian group.

So $(\mathbb{Z}, +, \cdot)$ is a ring.

Remark: an operation $$ is compatible with the equivalence relation $\asymp$ if the following holds: $x \asymp x' \wedge y \asymp y' \implies x * y \asymp x' * y'$*

Rational numbers

In the realm of integers, among the nonzero elements only 1 and -1 have a multiplicative inverse; in other words, division generally cannot be performed.

Consider on $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ the relation $\sim$ for which $(m, n) \sim (m', n')$ if $mn' = nm'$. And take the addition $(m, n) + (m', n') = (mn' + nm', nn')$ and the multiplication $(m, n) \cdot (m', n') = (mm', nn')$. The relation $\sim$ is an equivalence relation; we denote the set of equivalence classes by $\mathbb{Q}$. We call the elements of $\mathbb{Q}$ rational numbers.

$(\mathbb{Q}, +, \cdot)$ is an ordered field.

Real numbers

There is no number $a \in \mathbb{Q}$ whose square is 2. So not every number can be written in the form m/n ($m, n \in \mathbb{N}^+$).

Archimedean ordering:

An ordered field F is Archimedean ordered if $x, y \in F : \exists n \in \mathbb{N} : nx \geq y \quad (x > 0)$

The ordered field of rational numbers is Archimedean ordered, but does not have the upper bound property.

An ordered field with the upper bound property is called the field of real numbers, and is denoted $\mathbb{R}$. ($\exists! \mathbb{R}$)

1.4.2 Complex numbers

The need for complex numbers arose from the Cardano formula for solving cubic equations. Namely, in the case when the equation has three distinct real roots, a negative number appears under the root sign in the formula. The rules for computing with “imaginary” numbers, and the connection with trigonometric functions, were gradually clarified.

Definition

The set of complex numbers is $\mathbb{C} = \mathbb{R} \times \mathbb{R}$. $\mathbb{C}$ is a field with the addition $(x, y) + (x', y') = (x + x', y + y')$ and the multiplication $(x, y) \cdot (x', y') = (xx' - yy', y'x + xy')$. The set of complex numbers is not an ordered

field, since (by a theorem) in an ordered integral domain $x \neq 0 \Rightarrow x^2 > 0$. (This, however, does not hold for $(0, 1)^2 = i^2 = -1$).

[In the realm of complex numbers $(0, 0)$ is the zero element, $(1, 0)$ the identity element, the additive inverse of (x, y) is $(-x, -y)$, and the multiplicative inverse of the pair $(0, 0) \neq (x, y)$ is the pair $(\frac{x}{x^2+y^2}, \frac{-y}{x^2+y^2})$.]

Identification of real numbers

Since $(x, 0) + (x', 0) = (x + x', 0)$ and $(x, 0) \cdot (x', 0) = (xx', 0)$, we can identify all complex numbers $(x, 0)$, $x \in \mathbb{R}$ with $\mathbb{R}$.

Algebraic form of complex numbers

Since

$$(x, y) = (x, 0) + (y, 0) \cdot i = x + yi$$

we can also write complex numbers in the algebraic form $a + bi$.

Then the real number $\operatorname{Re}(z) = x$ is called the real part of the complex number $z = (x, y)$, and the real number $\operatorname{Im}(z) = y$ its imaginary part.

Conjugate

The conjugate of the complex number $z = x + yi$: $\bar{z} = x - yi$

Its properties:

1. $\overline{z + w} = \bar{z} + \bar{w}$
2. $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$
3. $\overline{\bar{z}} = z$
4. $z + \bar{z} = 2\operatorname{Re}(z)$
5. $z - \bar{z} = i \cdot 2\operatorname{Im}(z)$

Absolute value

The absolute value of the complex number $z = (x, y)$: $|z| = \sqrt{x^2 + y^2}$

Its properties:

1. $z \cdot \bar{z} = |z|^2$
2. $\frac{1}{z} = \frac{\bar{z}}{|z|^2}$
3. $|z| = \overline{|z|}$
4. $|z \cdot w| = |z| \cdot |w|$
5. $|z + w| \leq |z| + |w|$

Trigonometric form

- Argument

For $z \neq 0$ the argument of z is any $t \in \mathbb{R}$ for which $\operatorname{Re}(z) = |z|\cos(t)$, and $\operatorname{Im}(z) = |z|\sin(t)$. In other words the argument of z is the angle between the vector pointing from the origin to z and the positive real axis.

- Trigonometric form

The trigonometric form of the complex number z : $z = |z|(\cos(t) + i \cdot \sin(t))$

- De Moivre identities

Let $z = |z|(\cos(t) + i \cdot \sin(t))$, and $w = |w|(\cos(s) + i \cdot \sin(s))$. Then

$$z \cdot w = |z||w|(\cos(t + s) + i \cdot \sin(t + s))$$

$$\frac{z}{w} = \frac{|z|}{|w|}(\cos(t - s) + i \cdot \sin(t - s)) \quad (w \neq 0)$$

$$z^n = |z|^n(\cos(nt) + i \cdot \sin(nt)) \quad (n \in \mathbb{Z})$$

- Root extraction

Let $z^n = w$, then:

$$\sqrt[n]{w} = \left\{ z_k = \sqrt[n]{|w|} \left(\cos\left(\frac{t + 2k\pi}{n}\right) + i \sin\left(\frac{t + 2k\pi}{n}\right) \right), k = 0, \dots, n-1 \right\}$$

But since this notation can be confused with the (made unambiguous) real root extraction among reals, we do not use this notation. Instead, let us introduce the concept of the n -th complex root of unity:

$$\varepsilon_k = \cos\left(\frac{2k\pi}{n}\right) + i \cdot \sin\left(\frac{2k\pi}{n}\right), \quad k = 0, \dots, n-1$$

After this we can obtain the roots of w with the help of z and the n -th complex roots of unity:

$$z\varepsilon_0, \dots, z\varepsilon_{n-1}$$

1.5 Enumerations on finite sets

Finite sets

- Equivalence of sets

Sets X, Y are equivalent if there exists a bijection mapping X onto Y .

Notation: $X \sim Y$

- Finite and infinite sets

A set X is finite if $\exists n \in \mathbb{N} : X \sim \{1, 2, \dots, n\}$, otherwise infinite. If such an n exists, then it is unique, and then it is called the cardinality of the set. Notation: $\#(X)$

Pigeonhole principle

If X, Y are finite sets and $\#(X) > \#(Y)$, then a map $f : X \rightarrow Y$ cannot be one-to-one (that is, a bijection).

Enumerations

- Permutation

A permutation of a set A is a one-to-one map of it onto itself. The number of all permutations of the set A :

$$P_n = \prod_{k=1}^n k = n!$$

- Variation

The sequences $a_1, a_2, \dots, a_k$ of distinct elements that can be formed from the elements of the set A are called the variations of class k of the set A . If A is finite ($\#(A) = n$), then the number V_n^k equals the number of one-to-one maps from $\{1, 2, \dots, k\}$ into $\{1, 2, \dots, n\}$:

$$V_n^k = \frac{n!}{(n-k)!}$$

- Combination

The subsets of $k \in \mathbb{N}$ elements of a set A are called the combinations of class k . If A is finite, then the number C_n^k equals the number of k -element subsets of $\{1, 2, \dots, n\}$.

$$C_n^k = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- Permutation with repetition

The repetitions of the elements of the set $A = \{a_1, \dots, a_r\}$ are $i_1, \dots, i_r$. (The permutations with repetition of the elements are sequences of $i_1 + \dots + i_r = n$ terms, in which the element a_j occurs i_j

times.)

$$P_n^{i_1, \dots, i_r} = \frac{n!}{i_1! i_2! \dots i_r!}$$

- Variation with repetition

The (not necessarily distinct) sequences $a_1, \dots, a_k$ that can be formed from the elements of the finite set A are called the variations with repetition of class k of the set A .

$${}_iV_n^k = n^k$$

- Combination with repetition

Let A be a finite set. Selecting k elements from the set, allowing repetitions but disregarding order, we obtain the combinations with repetition of class k of the set A .

$${}_iC_n^k = \binom{n+k-1}{k}$$

Theorems

- Binomial theorem

$x, y \in R$ (commutative ring with identity), $n \in \mathbb{R}$. Then

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

- Multinomial theorem

$r, n \in \mathbb{N}$ and $x_1, x_2, \dots, x_r \in R$ (commutative ring with identity), then

$$(x_1 + \dots + x_r)^n = \sum_{i_1 + \dots + i_r = n} P_n^{i_1, \dots, i_r} x_1^{i_1} x_2^{i_2} \dots x_r^{i_r} \quad (i_1, \dots, i_r \in \mathbb{N})$$

- Inclusion-exclusion (sieve) formula

$X_1, \dots, X_k \subset X$ (finite set). f a function defined on X , mapping into an Abelian group. Let:

$$S = \sum_{x \in X} f(x)$$

$$S_r = \sum_{1 \leq i_1 \leq \dots \leq i_r \leq k} \left(\sum_{x \in X_{i_1} \cap \dots \cap X_{i_r}} f(x) \right)$$

and

$$S_0 = \sum_{x \in X \setminus \bigcup_{i=1}^k X_i} f(x)$$

Then

$$S_0 = S - S_1 + S_2 - S_3 + \dots + (-1)^k S_k$$

1.6 Basic concepts of number theory, division with remainder, linear congruence equations

1.6.1 Basic concepts of number theory

Divisibility in an integral domain with identity

R an integral domain with identity, $a, b \in R$. If $\exists c \in R : a = bc$, then b is a divisor of a (a is a multiple of b).

Notation: $b|a$

Except for $b = 0$, at most one such c exists.

Properties of divisibility in an integral domain with identity.

- If $b|a$ and $b'|a'$, then $bb'|aa'$
- $\forall a \in R : a|0$ (every element divides zero)
- $0|a \Leftrightarrow a = 0$ (zero divides only itself)
- $\forall a \in R : 1|a$ (the identity element divides every element)
- $b|a \Rightarrow \forall c \in R : bc|ac$
- $bc|ac$ and $c \neq 0 \Rightarrow b|a$
- $b|a_i$ and $c_i \in R, (i = 1, \dots, j) \Rightarrow b|\sum_{i=1}^j a_i c_i$
- the $|$ relation is reflexive and transitive

Irreducible element and prime element

$0, 1 \neq a \in R$ is irreducible if, for $a = bc$, b or c is a unit ($b, c \in R$).

$0, 1 \neq p \in R$ is prime if $\forall a, b \in R : p|ab$ implies $p|a$ or $p|b$

Greatest common divisor, least common multiple, relatively prime

R an integral domain with identity. $b \in R$ is a greatest common divisor of the elements $a_1, \dots, a_n \in R$ if $b|a_i$ and $(b'|a_i$ implies $b'|b$). If b is a unit, then $a_1, \dots, a_n$ are relatively prime.

$b \in R$ is a least common multiple of the elements $a_1, \dots, a_n \in R$ if $a_i|b$ and $(a_i|b'$ implies $b|b'$).

Extended Euclidean algorithm

The procedure determines the greatest common divisor ($d \in \mathbb{Z}$) of the numbers $a, b \in \mathbb{Z}$, as well as numbers $x, y \in \mathbb{Z}$ such that $d = ax + by$

Fundamental theorem of arithmetic

Every positive natural number can be uniquely (up to order) decomposed as a product of prime numbers.

Sieve of Eratosthenes

To determine the primes up to a given n : Write down the numbers from 2 to n . The first number (2) is prime, all its multiples are composite, cross these out. Among the remaining numbers, the first (3) is likewise prime, etc. At the end of the procedure the primes not greater than n remain.

1.6.2 Division with remainder

Let R be an integral domain with identity, $f, g \in R[x], g \neq 0$ and suppose that the leading coefficient of g is a unit in R . Then

$$\exists! q, r \in R[x] : f = g \cdot q + r \quad (\deg(r) < \deg(g))$$

1.6.3 Horner's scheme

Horner's method is suitable for computing the value of a polynomial at a substitution point. (Together with this, of course, one can also decide whether a given value c is a root of the polynomial or not. Above degree 4 there is not even an analytic solution for this.)

The essence of the method is that computing the value of the polynomial $f_n x^n + f_{n-1} x^{n-1} + \dots + f_0$, which would otherwise require a great many multiplications and additions. By transforming the polynomial, however, we can reduce the number of operations. Applying division with remainder:

$$f_n x^n + f_{n-1} x^{n-1} + \dots + f_0 = (f_n x^{n-1} + f_{n-1} x^{n-2} + \dots) x + f_0$$

Continuing this recursively we arrive at the following form:

$$(((f_n x + f_{n-1})x + f_{n-2})x + \dots)x + f_0$$

The computation of the substitution value can be carried out more easily in a table.

	f_n	f_{n-1}	f_{n-2}	$\cdots$	f_0
c	f_n	$f_n c + f_{n-1}$	$(f_n c + f_{n-1})c + f_{n-2}$	$\cdots$	$f(c)$

Filling in the table proceeds as follows:

1. In the first row we write the coefficients of the polynomial
2. In the first cell of the second row we write the value of the argument.
3. Under the leading coefficient we write it itself.
4. We continue by filling in the cells of the second row
5. We multiply the element of the previous cell by the argument
6. To the product we add the current coefficient
7. We write the sum into the current cell
8. We continue with step 5 until we reach the last cell

The substitution value of the polynomial goes into the last cell. (If this is zero, then the argument is a root of the polynomial.)

1.6.4 Linear congruence equations

Congruence

If $a, b, m \in \mathbb{Z}$ and $m|(a - b)$, then we say that a and b are congruent modulo m (Notation: $a \equiv b \pmod{m}$).

Congruence is an equivalence relation for any m . If $a \in \mathbb{Z}$ then the elements of the equivalence class are of the form $a + km, k \in \mathbb{Z}$.

Residue classes

We call the equivalence classes with respect to the modulus $m \in \mathbb{Z}$ residue classes. We represent residue classes by their elements. (The residue class represented by the element a is $\tilde{a} \pmod{m}$).

If some element of a residue class is relatively prime to the modulus, then all of them are, and the residue class is called a reduced residue class.

A system of pairwise incongruent integers is called a residue system.

If a residue system contains an element from every residue class, then it is a complete residue system.

If a residue system contains an element exactly from the reduced residue classes, then it is a reduced residue system.

Euler's φ function

$m > 0$ an integer. Euler's $\varphi(m)$ function gives the number of reduced residue classes modulo m . This obviously equals the number, among the numbers $0, 1, \dots, m-1$, of those relatively prime to m .

Euler–Fermat theorem

$m > 1$ an integer, a relatively prime to m , then:

$$a^{\varphi(m)} \equiv 1 \pmod{m}$$

Fermat's theorem

Let p be prime, and $a \in \mathbb{Z} : p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}$$

Solving a linear congruence

We look for the solutions of the congruence $ax \equiv b \pmod{m}$ ($a, b, m \in \mathbb{Z}$ known). This is equivalent to looking for an x for which (with some y) $ax + my = b$.

Let $d = \gcd(a, m)$. Since d divides $ax + my$, it must also divide b , otherwise there is no solution. Thus $\frac{a}{d}x + \frac{m}{d}y = \frac{b}{d}$. Then $a'x + m'y = 1$. With the help of the extended Euclidean algorithm we obtain numbers u, v with which $a'u + m'v = 1$ (since a', m' are relatively prime). Multiplying the equation by b' : $a'ub' + m'vb' = b' \Rightarrow x \equiv ub' \pmod{m'}$

Solving a system of linear congruences

For two linear congruences the solutions are $x \equiv a \pmod{m}$ and $x \equiv b \pmod{n}$. For the common solution we have to solve the equation $x = a + my = b + nz \Leftrightarrow my - nz = b - a$. There is a solution if and only if $d = \gcd(m, n)$ divides $b - a$. Then the solution can be written, with some integer x_1 , in the form $x \equiv x_1 \pmod{\text{lcm}(m, n)}$. (For more congruences the procedure can be continued.)

Chinese remainder theorem

$1 < m_1, \dots, m_n \in \mathbb{N}$ pairwise relatively prime, and $c_1, \dots, c_n \in \mathbb{Z}$. The system of congruences $x \equiv c_j \pmod{m_j}$ ($j = 1, \dots, n$) is solvable, and any two of its solutions are congruent $\pmod{m_1 m_2 \dots m_n}$

2 Polynomials and their operations

Definition

Let R be a ring. We regard a polynomial as a finite sum of the form $\sum_{i=0}^n f_i x^i$, where $n \in \mathbb{N}$, $f_i \in R$.

We call the term f_n the leading coefficient of the polynomial.

Operations

Let $R[x]$ be the ring over the infinite sequences $f = (f_0, f_1, \dots)$ (the ring of polynomials), where $f_i \in R$. Then the operations in $R[x]$:

- Addition:

$$f + g = (f_0 + g_0, f_1 + g_1, \dots) \quad (f, g \in R[x])$$

- Multiplication:

$$f \cdot g = h = (h_0, h_1, \dots) \quad (f, g, h \in R[x]), \text{ where}$$

$$h_k = \sum_{i+j=k} f_i g_j$$

Remark: If R is commutative, then so is $R[x]$. If R has identity with the identity element 1, then so does $R[x]$, with the identity element $(1, 0, 0, \dots)$.

3 Graphs

3.1 General and planar graphs

Basic concepts

- Undirected graph

An undirected graph is the ordered triple $G = (V, E, \varphi)$, where:

V - the set of vertices

E - the set of edges

φ - incidence relation ($\varphi \in E \times V$)

If $v \in \varphi(e)$, then v is incident to the edge e . ($v \in V, e \in E$). An edge always has two ends

- Edge and vertex types
 - Isolated vertex
 $v \in V$ is an isolated vertex if $\nexists e \in E : v \in \varphi(e)$
 - Parallel edge
 edges $e, e' \in E$ are parallel edges if $\varphi(e) = \varphi(e')$
 - Loop edge
 $e \in E$ is a loop edge if $|\varphi(e)| = 1$
- Directed graph
 A directed graph is the ordered triple $G = (V, E, \psi)$, where:
 V - the set of vertices
 E - the set of edges
 ψ - incidence relation ($\psi \in E \rightarrow V \times V$)
 $\psi(e) = (v, v')$, where v is the start point of the edge e , v' its end point.

Finite, simple graphs - basic concepts

- Simple graph
 a graph G is simple if it does not contain parallel or loop edges
- Finite graph a graph $G = (V, E, \varphi)$ is finite if V, E are finite sets.
- Adjacency, degree
 Two edges are adjacent if they have a common point.
 Two vertices are adjacent if they have a common edge.
 The number of neighbors of $v \in V$ is the degree of v . [Notation: $\deg(v) = d(v)$]
- r -regular graphs
 a graph G is r -regular if every vertex has degree r
- Complete graph
 a graph G is a complete graph if every edge is drawn in, in other words it is $(|V| - 1)$ -regular. (Notation: $K_{|V|}$)
- Bipartite graph
 G is a bipartite graph if $V = V' \cup V''$ and $V' \cap V'' = \emptyset$ (disjoint), and an edge runs only between V' and V'' .
If, moreover, every edge between V' and V'' is drawn in this way, then it is a complete bipartite graph. (Notation: $K_{n,m}$, where $n = |V'|, m = |V''|$)
- Subgraph
 $G = (V, E, \varphi)$ is a subgraph of $G' = (V', E', \varphi')$ if $V \subset V' \wedge E \subset E' \wedge \varphi \subset \varphi'$
- Walk, trail, path
 In a graph G a walk of length n from v to v' is a sequence

$$v_0, e_1, v_1, \dots, v_{n-1}, e_n, v_n$$
 for which $v = v_1, v' = v_n$ and $v_{i-1}, v_i \in \varphi(e_i)$
 A walk is a trail if every edge appears at most once in the sequence.
 A trail is a path if every vertex appears at most once in the sequence.
 A walk/trail/path is closed if its start and end points coincide, otherwise open.
- Connected graph A graph is connected if there is a path between any two vertices.
This relation is an equivalence relation, whose equivalence classes are called components.

- Labeled, weighted graph

$G = (V, E, \varphi, C_e, c_e, C_v, c_v)$ ordered 7-tuple denotes a labeled graph, where C_e, C_v are arbitrary sets, and

$$c_e : E \rightarrow C_e$$

$$c_v : E \rightarrow C_v$$

If $C_e = C_v = \mathbb{R}^+$, then the graph is called a weighted graph, and w is the weight of the vertex/edge.
 $(w(e) = c_e(e), w(v) = c_v(v))$

Planarity

Concepts

- Planarity

A graph is planar if it can be drawn so that its edges do not cross each other.

- Topological isomorphism

Two graphs are topologically isomorphic if by finitely many repetitions of the following step or its reverse we obtain one from the other: We remove a degree-two vertex and connect its neighbors.

- Region

If a graph G is planar, then the regions are the plane figures bounded by the edges. (The unbounded plane figure is also a region.)

Theorems

1. Every finite graph can be drawn in $\mathbb{R}^3$.
2. A finite graph is planar $\iff$ it can be drawn on a sphere
3. Euler's theorem:
If the finite graph G is a connected, planar graph, then:

$$|E| + 2 = |V| + |T|$$

4. Kuratowski's theorem:

A finite graph is planar if and only if it does not contain a subgraph topologically isomorphic to K_5 or $K_{3,3}$.

3.2 Trees

Tree

A graph is called a tree if it is connected and acyclic.

Spanning tree

F is a subgraph of G . If F is a tree and its vertex set equals the vertex set of G , then F is called a spanning tree of G .

Theorems

- If G is a simple graph, then the following conditions are equivalent:
 1. G is a tree
 2. G is connected, but deleting any edge it no longer is
 3. Between two distinct vertices there is only one path
 4. G is acyclic, but adding an edge it no longer is
- If G is a simple finite graph, then the following conditions are equivalent:
 1. G is a tree

2. G has no cycle and has $n - 1$ edges
3. G is connected and has $n - 1$ edges

Directed tree

A tree for which: $\exists v \in V : d^-(v) = 0$ and $\forall v' \neq v : d^-(v') = 1$ (One vertex has in-degree 0, the others 1)

Further concepts:

- the vertex $r \in V, d^-(r) = 0$ is called the root
- the level of vertex v' is the length of the path r, v'
- $(v, v') \in \psi(e)$, v is the parent of v' , v' the child of v .
- v is a leaf if $d^+(v) = 0$

3.3 Eulerian and Hamiltonian graphs

3.3.1 Eulerian graph

Eulerian trail

An Eulerian trail is a trail from v to v' in the graph in which every edge appears. If $v = v'$ then this trail is also usually called an Eulerian circuit. A graph having an Eulerian trail is called an Eulerian graph.

Theorem

In a connected finite graph an Eulerian circuit exists if and only if every vertex has even degree.

3.3.2 Hamiltonian graph

A Hamiltonian path is a path from v to v' in the graph that contains every vertex. If $v = v'$ then this path is also usually called a Hamiltonian cycle. A graph having a Hamiltonian path is called a Hamiltonian graph.

3.4 Data structures of graphs

For the computer representation of graphs, linked lists or matrices are most often used. Linked lists are more economical for sparse graphs, while matrices are economical for dense graphs.

Incidence matrix

For a directed graph $G = (V, E, \psi)$ we can represent the graph with the help of a matrix $A = \{0, 1, -1\}^{n \times m}$, where $V = \{v_1, \dots, v_n\}$, and $E = \{e_1, \dots, e_m\}$. Then the individual elements of the matrix:

$$a_{ij} = \begin{cases} 1 & \text{if } v_i \text{ is the start point of } e_j \\ -1 & \text{if } v_i \text{ is the end point of } e_j \\ 0 & \text{otherwise} \end{cases}$$

If G is undirected, then $a_{ij} = |a_{i,j}|$

Vertex matrix (adjacency matrix)

With the above notation, in the directed case $B \in \mathbb{Z}^{n \times n}$, where b_{ij} denotes the number of edges going from v_i to v_j .

If G is undirected, then b_{ii} is the number of loop edges of v_i , otherwise b_{ij} is the number of edges between the vertices v_i and v_j .